



National Technical University of Athens

School of Electrical and Computer Engineering

9th Semester

Renewable Energy Sources (RES)

Semester Project

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PROBLEM 1: WIND TURBINE DESIGN

Wind Turbine Data

In this problem, a three-bladed variable-speed and variable-pitch wind turbine is considered. The wind turbine is equipped with a synchronous generator and power converters on the stator side. The main geometric and operational characteristics of the wind turbine are given in Table 1.

Rated rotational speed	11 rpm
Rotor diameter	136 m
Hub height	120 m
Gearbox ratio	1:40
Cut-in wind speed ($V_{\text{cut-in}}$)	3 m/s
Cut-out wind speed ($V_{\text{cut-out}}$)	26 m/s
Pitch angle range	0° - 40°
Number of poles	12
Generator and power converter efficiency	96 %
Rated AC voltage of the grid-side converter	400 V

Table 1: Characteristics of the wind turbine

The aerodynamic power coefficient C_p of the blades is approximated by the following relation:

$$C_p(\lambda, \beta) = 0.5176 \left(\frac{116}{\lambda_i} - 0.4\beta - 5 \right) \cdot e^{\frac{-21}{\lambda_i}} + 0.0068\lambda \quad (1)$$

where β is the pitch angle of the rotor blades (in degrees) and λ is the tip-speed ratio.

The intermediate variable λ_i is defined as:

$$\frac{1}{\lambda_i} = \frac{1}{\lambda + 0.08\beta} - \frac{0.035}{\beta^3 + 1} \quad (2)$$

Rotational losses are given by:

$$P_{\text{rot,loss}}(\text{kW}) = -5 + 7N_R \quad (3)$$

where N_R (rpm) is the rotor rotational speed. The air density is assumed to be $\rho = 1.225 \text{ kg/m}^3$ and the wind shear exponent is $\alpha = 0.15$.

1. Generator-Side Voltage Frequency as a Function of Rotor Speed

The frequency of the voltage on the generator side is determined by the rotational speed of the synchronous generator and the number of pole pairs. For a wind turbine with gearbox ratio $G = 40$ and a generator with $P = 12$ poles (i.e., $p = 6$ pole pairs), the generator rotational speed N_g is related to the rotor rotational speed N_R by:

$$N_g = G N_R \quad (4)$$

The corresponding electrical frequency of the generated voltage on the generator side is:

$$f_{\text{gen}} = \frac{p N_g}{60} = \frac{p G N_R}{60} = \frac{6 \cdot 40}{60} N_R = 4N_R \quad (5)$$

For wind turbine operation below rated conditions, maximum power point tracking (MPPT) control is assumed, with a pitch angle of $\beta = 0^\circ$. Figure 1 shows the variation of the power coefficient C_p as a function of the tip-speed ratio λ .

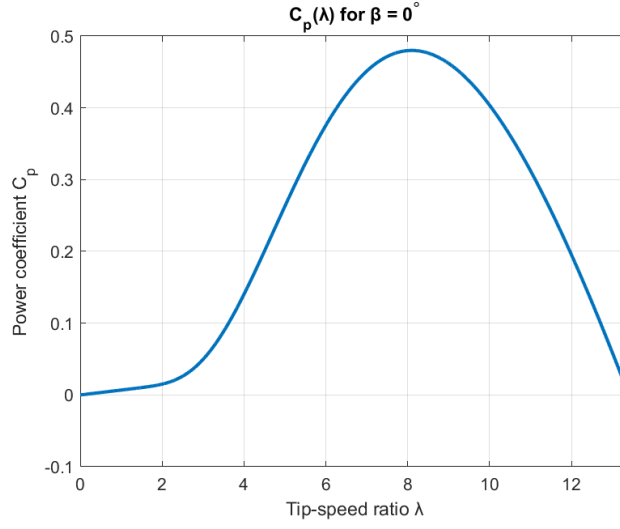


Figure 1: Variation of power coefficient C_p as a function of tip-speed ratio λ for $\beta = 0^\circ$

From the above plot, the optimal tip-speed ratio λ_{opt} that maximizes the captured power is obtained together with the corresponding maximum value $C_{p,\text{max}}$:

$$\lambda_{\text{opt}} = 8.1 \quad (6)$$

$$C_{p,\text{max}} = 0.48 \quad (7)$$

Based on the optimal value λ_{opt} , the rotor rotational speed at cut-in can be calculated by considering the wind speed $V_{\text{cut-in}} = 3$ m/s. For wind speeds below the cut-in speed, the wind turbine remains disconnected from the grid and does not develop rotational motion, resulting in a zero generator-side frequency, $f_{\text{gen}} = 0$. The rotor rotational speed $N_{R,\text{cut-in}}$ at which voltage generation to the grid begins is calculated below.

The tip-speed ratio is defined as:

$$\lambda = \frac{\omega_R R}{V_w} \quad (8)$$

Therefore, under MPPT operation, $\lambda = \lambda_{\text{opt}}$ and for the cut-in wind speed $v = V_{\text{cut-in}}$:

$$\omega_R = \frac{\lambda_{\text{opt}} V_{\text{cut-in}}}{R} \implies N_{R,\text{cut-in}} = \frac{60}{2\pi} \frac{\lambda_{\text{opt}}}{R} V_{\text{cut-in}} = \frac{60 \cdot 8.1 \cdot 3}{2\pi \cdot 68} = 3.41 \text{ rpm} \quad (9)$$

The final characteristic curve of the generator-side voltage frequency as a function of rotor rotational speed is shown in Figure 2. It can be observed that for $N_R < N_{R,\text{cut-in}}$ the frequency remains zero, whereas for higher rotational speeds it increases linearly.

At the rated rotor speed $N_R = 11$ rpm, the maximum generator-side voltage frequency is:

$$f_{\text{gen,max}} = 44 \text{ Hz} \quad (10)$$

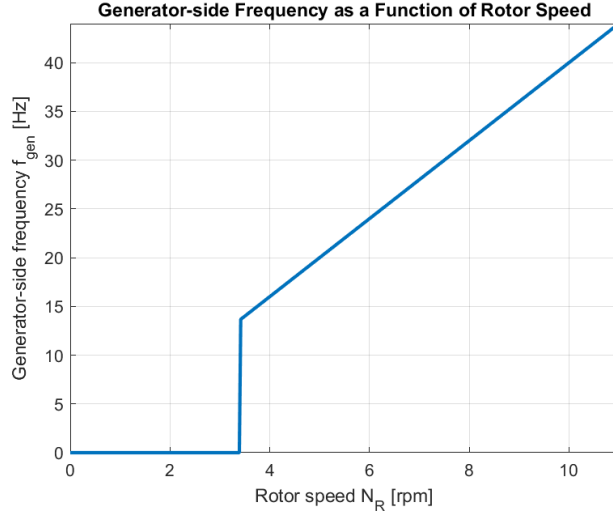


Figure 2: Generator-side frequency f_{gen} as a function of rotor speed N_R

2. Output Power–Rotor Speed Characteristic ($P_{out} - N_R$)

For the design of the output power characteristic as a function of rotor speed, variable-speed operation below rated conditions is considered using maximum power point tracking (MPPT), with $\beta = 0^\circ$ and $\lambda = \lambda_{opt} = 8.1$.

The wind power is:

$$P_{wind} = \frac{1}{2} C_{p,max} \rho A V_w^3 \quad (11)$$

Combining this expression with equation (8), we obtain:

$$P_{wind} = \frac{1}{2} C_{p,max} \rho A \cdot \left(\frac{2\pi}{60} \frac{R}{\lambda_{opt}} N_R \right)^3 \quad (12)$$

Rotational losses are given by:

$$P_{rot,loss}(\text{kW}) = -5 + 7N_R, \quad (13)$$

Thus, the output power is obtained by subtracting the rotational losses from the mechanical power and multiplying by the overall efficiency of the generator and power converters:

$$\begin{aligned} P_{out} &= \eta (P_{wind} - P_{rot,loss}) \\ &= 0.96 \cdot \left(\frac{1}{2} \cdot 0.48 \cdot 1.225 \cdot \pi \cdot 68^2 \cdot \left(\frac{2\pi \cdot 68}{60 \cdot 8.1} \cdot N_R \right)^3 - 1000 \cdot (-5 + 7N_R) \right) \\ &= 2785.76 N_R^3 - 6720 N_R - 4800 \end{aligned} \quad (14)$$

The cut-in wind speed V_{cut-in} is also taken into account, below which the wind turbine is considered disconnected and the output power is zero. Therefore, for $N_R < N_{R,cut-in}$, $P_{out} = 0$, while for $N_R \geq N_{R,cut-in}$ the power follows the calculation procedure described above. The resulting $P_{out} - N_R$ characteristic is shown in Figure 3.

The rated output power of the wind turbine corresponds to the maximum (rated) rotor speed $N_R = N_{R,nom}$. From the calculations, we obtain:

$$P_{out,nom} = 3.639 \text{ MW} \quad (15)$$

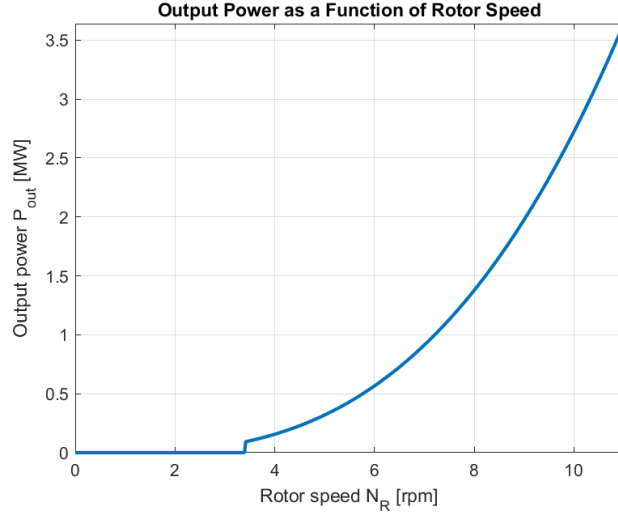


Figure 3: Output power characteristic P_{out} as a function of rotational speed N_R

3. Power Curve and Pitch Control Characteristic (β - V_w)

The operation of the wind turbine is divided into three main regions:

- Below the cut-in wind speed, $V_w < V_{cut-in}$, where $P_{out} = 0$ and $\beta = 0$.
- MPPT operation below rated conditions, $V_{cut-in} \leq V_w \leq V_{rated}$, where the power follows the operation described in the previous question based on equation (14) and $\beta = 0$.
- Pitch-control region, $V_{rated} < v_w \leq V_{cut-out}$, for which the following applies.

For wind speeds above rated, pitch control is applied so that the output power is maintained at the rated value $P_{out,rated}$. In this region, the rotor speed is assumed to remain constant at its rated value ($\omega_R = \omega_{rated}$), hence:

$$\omega_R = \omega_{rated} \Rightarrow \lambda = \frac{\omega_{rated} R}{V_w} \quad (16)$$

The required pitch angle β is calculated as follows:

$$P_{out} = P_{out,rated} \Rightarrow \eta \left(C_p(\lambda, \beta) \frac{1}{2} \rho A V_w^3 - P_{rot,loss} \right) = P_{out,nom} \quad (17)$$

Equivalently:

$$\begin{aligned} C_p(\lambda, \beta) &= \frac{P_{out,nom}/\eta + P_{rot,loss}}{\frac{1}{2} \rho A V_w^3} \\ &= \frac{\frac{P_{out,nom}}{0.96} + (7N_R - 5) \cdot 10^3}{\frac{1}{2} \cdot 1.225 \cdot (\pi \cdot 68^2) \cdot V_w^3} \\ &= \frac{1.0417 \cdot 3629000 + 7000N_R - 5000}{8895.3 \cdot V_w^3} \\ &= \frac{424.418 + 0.7869N_R}{V_w^3} \end{aligned} \quad (18)$$

Substituting into the above expression equations (1), (2) and (16) yields the pitch-angle characteristic, which increases with wind speed so that C_p decreases and the output power remains constant.

The final power curve as a function of wind speed is shown in Figure 4, while the corresponding pitch-angle control characteristic is shown in Figure 5.

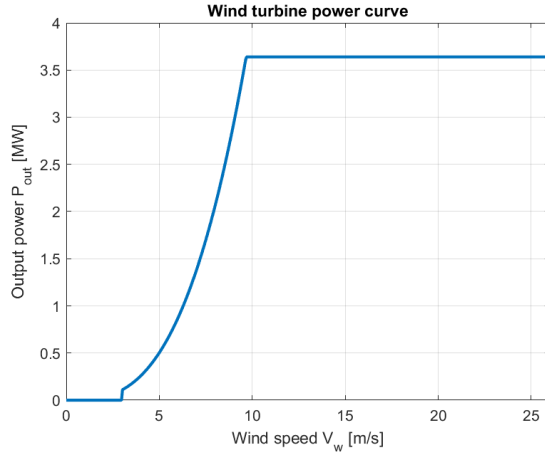


Figure 4: Output power curve P_{out} as a function of wind speed V_w

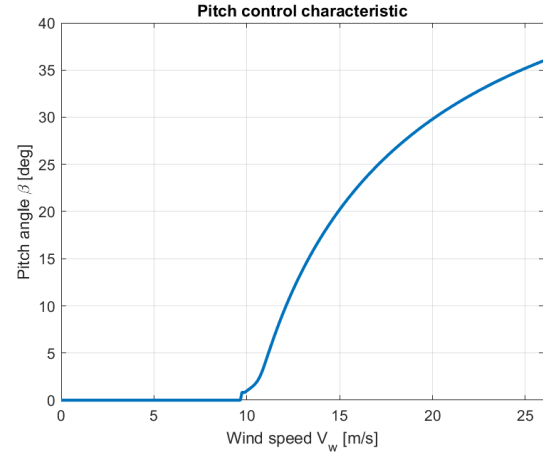


Figure 5: Pitch angle control characteristic β as a function of wind speed V_w

4. Calculation of Output Power, Rotor Speed, and Pitch Angle from Wind Measurements

Wind-speed measurements at a height of 10 m over a time interval of 10 minutes are given, as shown in Figure 6. The objective is, for the considered variable-speed and variable-pitch wind turbine, to calculate and plot the output power $P_{\text{out}}(t)$, the rotor rotational speed $N_R(t)$, and the pitch angle $\beta(t)$.

First, the measurements at the reference height $Z_{\text{ref}} = 10$ m are converted to wind speed at the hub height Z_{hub} using the wind-shear exponent α through:

$$V_{w,\text{hub}} = V_w \left(\frac{Z_{\text{hub}}}{Z_{\text{ref}}} \right)^\alpha = V_w \left(\frac{120}{10} \right)^{0.15} = 1.452 V_w \quad (19)$$

The time variation of wind speed at both 10 m and 120 m is shown in the same plot in Figure 6.

To calculate $P_{\text{out}}(t)$, $N_R(t)$, and $\beta(t)$, the steady-state characteristics obtained in Question 3 are used: the power curve $P_{\text{out}}(v)$ and the pitch-control characteristic $\beta(v)$. The time evolution is obtained by applying the corresponding wind speed value $V_w(t)$ to the above characteristics at each time instant t . The rotor-speed characteristic $N_R(t)$ is obtained using equation (8).

The resulting time variations of output power, rotor speed, and pitch angle are presented in Figures 7, 8, and 9, respectively.

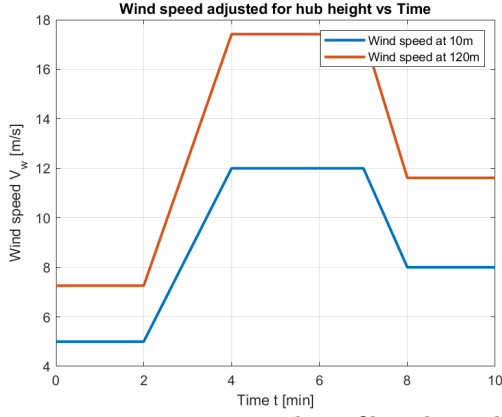


Figure 6: 10-minute wind profile adjusted for Z_{hub}

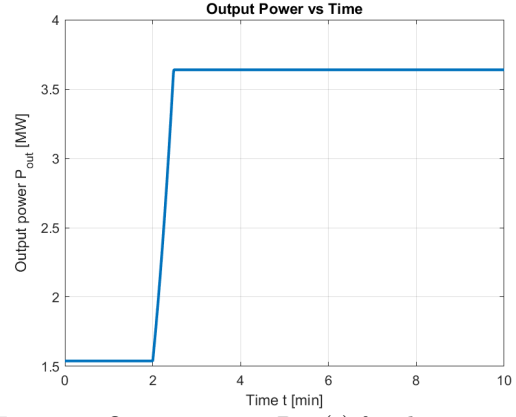


Figure 7: Output power $P_{out}(t)$ for the 10-minute wind profile

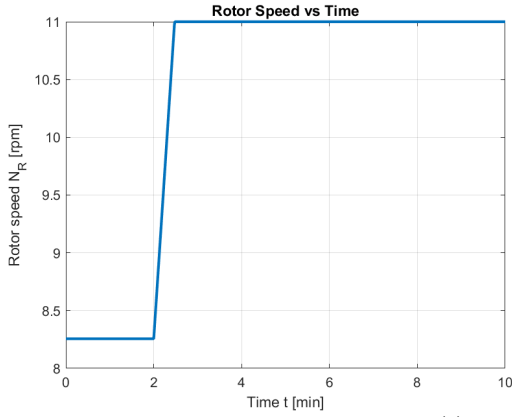


Figure 8: Rotor rotational speed $N_R(t)$ for the 10-minute wind profile

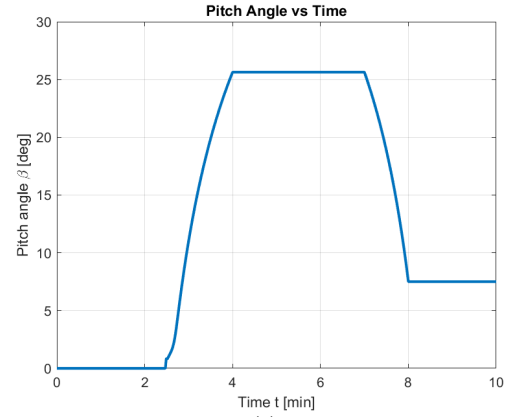


Figure 9: Pitch angle $\beta(t)$ for the 10-minute wind profile

5. Wind Turbine Response to a Frequency Disturbance

The considered wind turbine is required to satisfy the technical connection requirements of Commission Regulation (EU) 2016/631 regarding its participation in frequency regulation. In particular, a 5% droop and a frequency threshold of $f_{th} = 50.2$ Hz are assumed.

Following a disturbance in the Power System (PS) to which the wind turbine is connected, a frequency response is observed as shown in Figure 10. Before the event, the wind turbine output power is $P_{out,0} = 2$ MW and the wind speed is assumed to remain constant.

Based on these data, and assuming that the wind turbine operates in the MPPT region, the wind speed can be calculated using equation (14), yielding:

$$V_w = 7.9214 \text{ m/s} \quad (20)$$

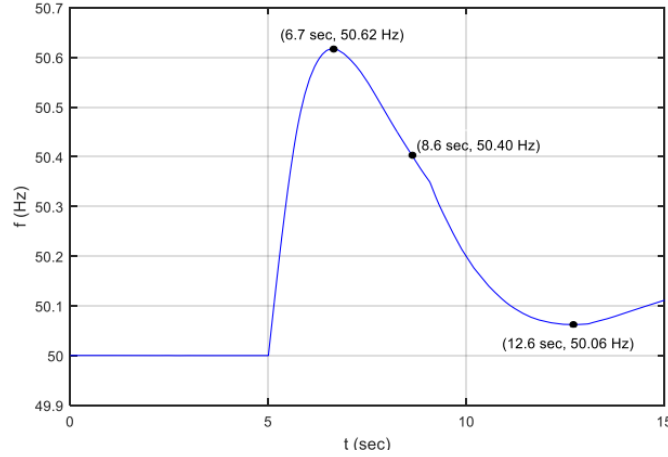


Figure 10: System Frequency Response during a disturbance

According to the regulation, when the frequency exceeds the threshold value f_{th} , the wind turbine must reduce its output power according to a droop characteristic, based on the following relation (for a 5% droop):

$$\frac{\Delta P}{P_{nom}} = \frac{1}{0.05} \frac{\Delta f}{f_{nom}} \quad (21)$$

The wind turbine output power at each time instant of Figure 10 is calculated as:

$$P_{out}(t) = P_{out,0} + \Delta P \quad (22)$$

Since the wind speed remains constant, the power variation is achieved through pitch-angle control β . Specifically, the required reduction in the aerodynamic power coefficient C_p satisfies:

$$C_p(\lambda, \beta) = \frac{P_{out,nom}/\eta + P_{rot,loss}}{\frac{1}{2}\rho A V_w^3} \quad (23)$$

The corresponding pitch angle $\beta(t)$ is obtained by inverting the relation $C_p(\lambda, \beta)$ so that the required output power is achieved at each time instant. At the same time, the rotor rotational speed is maintained constant, since there is no change in wind speed and regulation is performed exclusively through pitch control. We assume that the wind turbine was operating at λ_{opt} before the disturbance, so the rotor speed can be calculated from equation (8), as follows:

$$\omega_R = \frac{\lambda_{opt} \cdot V_w}{R} = \frac{8.1 \cdot 7.9214}{68} = 0.944 \implies N_R = 9.012 \text{ rpm} \quad (24)$$

Therefore, the results for the points of Figure 10 are presented in Table 2:

t [s]	f [Hz]	P_{out} [MW]	N_R [rpm]	β [°]
6.7	50.62	1.3891	9.012	5.1673
8.6	50.40	1.7091	9.012	1.4352
12.6	50.06	2	9.012	0

Table 2: Frequency Response Characteristic Results

It should be noted that the third operating point ($f = 50.06 \text{ Hz}$) is below the frequency threshold ($f_{th} = 50.2 \text{ Hz}$). Therefore, at this point the wind turbine returns to MPPT operation, with zero pitch angle and maximum power for the given wind speed.

6. Maximum Required Apparent Power of the Grid-Side Converter

Figure 11 defines an allowable operating region in the power plane, normalized with respect to the maximum active output power P_{nom} . Each point on the curve corresponds to a pair of values $(P/P_{nom}, Q/P_{nom})$.

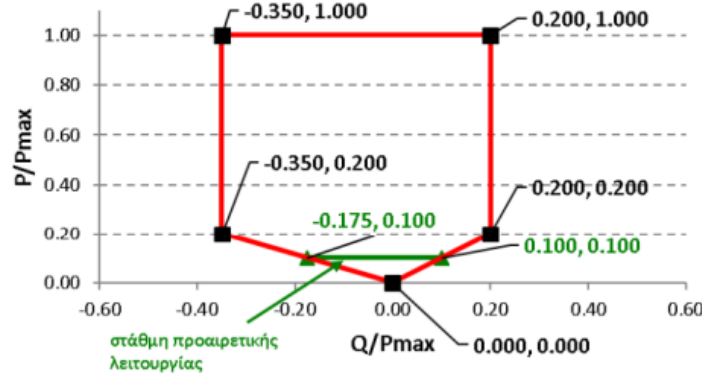


Figure 11: Reactive power capability requirement

The apparent power of the converter is given by:

$$S = \sqrt{P^2 + Q^2}. \quad (25)$$

In normalized form:

$$\frac{S}{P_{nom}} = \sqrt{\left(\frac{P}{P_{nom}}\right)^2 + \left(\frac{Q}{P_{nom}}\right)^2} \quad (26)$$

To determine the minimum required rated size of the grid-side converter, the maximum required apparent power of the wind turbine must be identified. We observe that at the point $(-0.350, 1.000)$ the maximum magnitude is reached, and from the previous question we have calculated $P_{nom} = 3.639$ MW.

Therefore, the required apparent power of the converter is:

$$S_{max} = P_{nom} \cdot \sqrt{\left(\frac{1}{P_{nom}}\right)^2 + \left(\frac{-0.35}{P_{nom}}\right)^2} = 3.639 \cdot \sqrt{1^2 + 0.35^2} = 3.855 \text{ MVA} \quad (27)$$

7. Reactive Current Control and Reactive Power Provision Due to Voltage Variation

This question examines the wind turbine response in terms of reactive current and reactive power during a voltage disturbance, according to the control characteristic shown in Figure 12 and the voltage response shown in Figure 13.

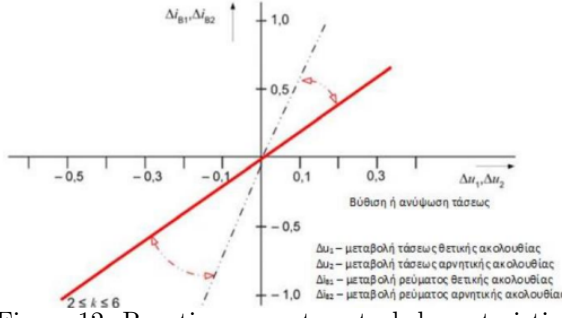


Figure 12: Reactive current control characteristic versus voltage deviation

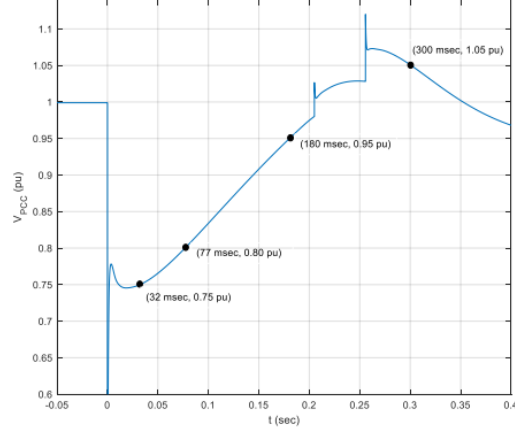


Figure 13: Voltage response at the Point of Common Coupling

The control characteristic relating voltage variation to reactive current is given by:

$$k = \frac{\frac{\Delta i_B}{i_n}}{\frac{\Delta u}{u_n}} = 2 \quad (28)$$

The above relation does not apply for $0.85 < u < 1.1$, which corresponds to the control deadband.

Assuming that the converter output voltage is approximated by the voltage at the Point of Common Coupling (PCC) and that the maximum apparent power is given by equation (27), the reactive power is calculated as:

$$Q = u \cdot \Delta i_B \cdot S_{\max} \quad (29)$$

At the same time, the active power capability is limited by the maximum apparent power:

$$P = \sqrt{S_{\max}^2 - Q^2} \quad (30)$$

The results for the four operating points are summarized in Table 3. It is worth noting that since the last two operating points have voltage deviations within the deadband, no reactive power is injected and the wind turbine operates at the rated power calculated in Question 2. Also, at the second operating point the maximum active power is calculated as $P_{\max} = 3.652$ MW, but since $P_{nom} = 3.639$ MW, the nominal value is selected.

u [pu]	Δi_B [pu]	Q [pu]	Q [MVar]	P [pu]	P [MW]
0.75	0.5	0.375	1.445	0.927	3.574
0.80	0.4	0.320	1.234	0.944	3.639
0.95	0	0	0	0.944	3.639
1.05	0	0	0	0.944	3.639

Table 3: Reactive current and power response to a voltage disturbance

8. Low Voltage Ride Through (LVRT)

The LVRT (Low Voltage Ride Through) curve is shown in Figure 14 and defines the minimum permissible voltage levels at the Point of Common Coupling (PCC) as a function of the duration of the disturbance, below which the wind turbine is or is not allowed to remain connected to the grid.

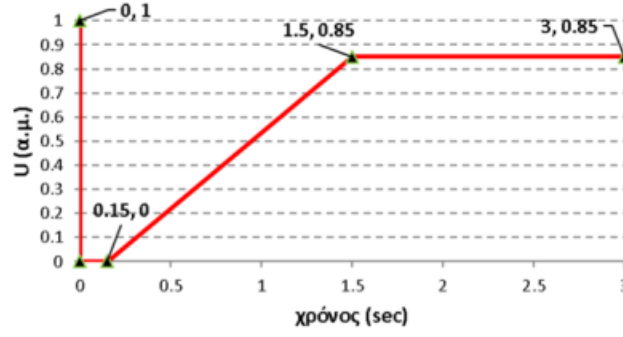


Figure 14: LVRT requirement

According to Figure 14, the LVRT curve is defined by three regions:

- For $0 < t < 0.15$: $U = 0 \text{ p.u.}$
- For $0.15 < t < 1.5$: $U = 0.6296 \cdot t - 0.0944 \text{ p.u.}$
- For $0.85 < t < 3$: $U = 0.85 \text{ p.u.}$

If the actual PCC voltage falls below the LVRT curve for a period longer than the permissible duration, the wind turbine is allowed to disconnect from the grid.

Check for Fault A

Figure 15 shows the PCC voltage response of the wind turbine during Fault A. It can be observed that immediately after the fault occurs, the voltage drops to approximately 0.2 p.u. and remains at this level for 0.4 sec. Based on the straight-line segment defining the LVRT curve, at $t = 0.4 \text{ sec}$ we obtain $U = 0.1574 \text{ p.u.}$, therefore the actual voltage remains above the system operator's requirement. The voltage then recovers and increases sharply, remaining within the permissible limits. Therefore, for Fault A, the wind turbine **must remain connected** to the grid throughout the fault.

Check for Fault B

Similarly, Figure 16 shows a similar PCC voltage response of the wind turbine during Fault B. It can be observed that immediately after the fault occurs, the voltage drops to approximately 0.4 p.u. and remains there for 0.4 sec. Since $U > 0.1574 \text{ p.u.}$, the voltage curve during this part of the fault remains above the system operator's requirement. The voltage then recovers and increases sharply, remaining within the permissible limits. Therefore, also for Fault B, the wind turbine **must remain connected** to the grid throughout the fault.

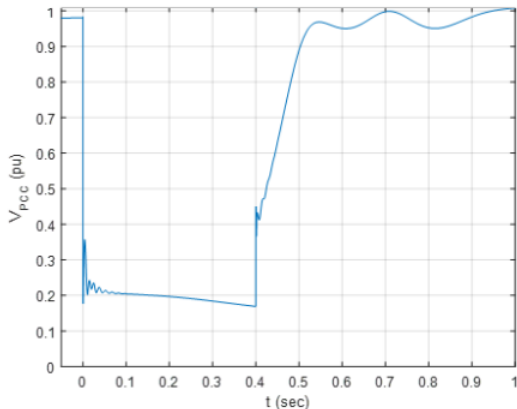


Figure 15: Voltage response during Fault A

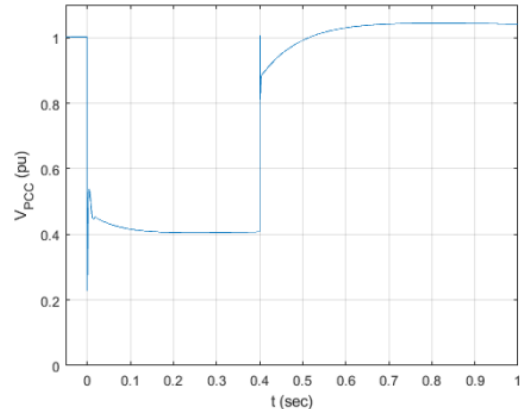


Figure 16: Voltage response during Fault B

PROBLEM 2: GRID CONNECTION

Grid Data - PV Plant

In this problem, a Medium-Voltage (MV) network with a nominal voltage of 20 kV is considered, to which photovoltaic (PV) plants and inductive loads are connected at successive nodes. The network is supplied by a substation with a constant voltage at the MV bus, while the distribution lines are implemented using ACSR-95 conductors with per-unit-length impedance $R = 0.22 \Omega/km$ and $X = 0.33 \Omega/km$.

A new PV plant with a rated power of 4 MW is planned to be connected at Node 3. The plant operates at unity power factor, but its maximum active power output is limited to 73% of the nominal value due to a ministerial decision. The main electrical characteristics of the network, the loads, and the PV plants are summarized in Figure 17. Network losses are assumed negligible, and a coincidence factor of 0.9 is considered for the already installed PV plants, together with a short-circuit current equal to 1.2 times the rated current.

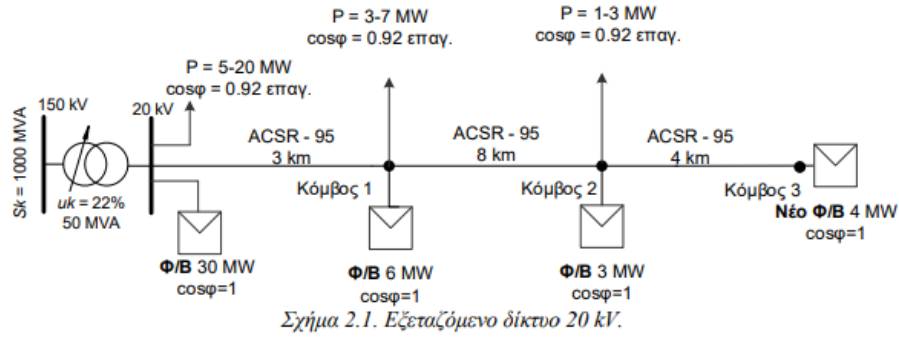


Figure 17: The 20 kV network under study

1. Transformer Thermal Limit Check

To check the transformer thermal limit, two different operating conditions of the network must be considered:

- Maximum power consumption, during which the PV plants generate no power and the loads operate at maximum consumption.
- Maximum power generation, during which the PV plants generate their nominal power and the loads consume the minimum possible power.

Maximum Power Consumption

The maximum load power is:

$$P_{L,\max} = P_{L,0,\max} + P_{L,1,\max} + P_{L,2,\max} = 20 + 7 + 3 = 30 \text{ MW} \quad (31)$$

The apparent power through the transformer is:

$$S_{\max,\text{import}} = \sqrt{P^2 + Q^2} = \frac{P}{\cos \varphi} = \frac{30}{0.92} = 32.6 \text{ MVA} < 50 \text{ MVA} \quad (32)$$

Maximum Power Generation

The minimum load power is:

$$P_{L,\min} = P_{L,0,\min} + P_{L,1,\min} + P_{L,2,\min} = 5 + 3 + 1 = 9 \text{ MW} \quad (33)$$

$$Q_{L,\min} = P_{L,\min} \cdot \tan \phi = 9 \cdot 0.4259 = 3.833 \text{ MVar} \quad (34)$$

The maximum PV generation power (where n is the coincidence factor) is:

$$P_{G,\max} = n \cdot (P_{G,0,\max} + P_{G,1,\max} + P_{G,2,\max} + P_{G,3,\max}) = 0.9 \cdot (30 + 6 + 3) + 4 \cdot 0.73 = 38.02 \text{ MW} \quad (35)$$

The apparent power through the transformer is:

$$S_{\max,\text{export}} = \sqrt{(P_{G,\max} - P_{L,\min})^2 + Q_{L,\min}^2} = \sqrt{(38.02 - 9)^2 + 3.833^2} = 29.27 \text{ MVA} < 50 \text{ MVA} \quad (36)$$

Therefore, the 50 MVA transformer is not overloaded, neither under maximum power import nor under maximum power export.

2. Short-Circuit Level Check at the 20 kV MV Bus

To determine the short-circuit level at the specified bus, the equivalent impedance of the system is calculated as follows.

For the 150 kV network with $S_{\max} = 1000 \text{ MVA}$:

$$Z_{k,\text{HV}} = \frac{V_{\text{HV}}^2}{S_{\max}} = \frac{(150 \text{ kV})^2}{1000 \text{ MVA}} = 22.5 \text{ } \Omega \quad (37)$$

Referred to MV (with voltage ratio 20/150):

$$Z_{k,\text{MV}} = Z_{k,\text{HV}} \left(\frac{20}{150} \right)^2 = 22.5 \cdot \left(\frac{20}{150} \right)^2 = 0.40 \text{ } \Omega \quad (38)$$

With $u_k = 22\%$ on a base of $S_n = 50 \text{ MVA}$, $V_{\text{LV}} = 20 \text{ kV}$:

$$Z_k = u_k \frac{V_{\text{LV}}^2}{S_n} = 0.22 \cdot \frac{(20 \text{ kV})^2}{50 \text{ MVA}} = 0.22 \cdot \frac{400}{50} = 1.76 \text{ } \Omega \quad (39)$$

Therefore:

$$Z_{k,\text{tot}} = Z_{k,\text{MV}} + Z_k = 0.40 + 1.76 = 2.16 \text{ } \Omega \quad (40)$$

To calculate the total short-circuit power at the bus, the transformer short-circuit contribution is added to the contribution of the PV plants and compared with the permissible short-circuit level at the 20 kV MV bus. The loads are neglected in the short-circuit calculation.

The transformer short-circuit power is:

$$S_{k,\text{MV}} = \frac{V_{\text{MV}}^2}{Z_{k,\text{tot}}} = \frac{(20 \text{ kV})^2}{2.16 \text{ } \Omega} = 185 \text{ MVA} \quad (41)$$

The PV short-circuit contribution is:

$$S_{k,G} = 1.2 \cdot (P_{G,0,\max} + P_{G,1,\max} + P_{G,2,\max} + P_{G,3,\max}) = 1.2 \cdot (30 + 6 + 3 + 4) = 51.6 \text{ MVA} \quad (42)$$

Hence:

$$S_{k,\text{tot}} = S_{k,MV} + S_{k,G} = 185 + 51.6 = 236.6 \text{ MVA} < 250 \text{ MVA} \quad (43)$$

Therefore, the calculated short-circuit level at the MV bus remains within the network design limits.

3. Check of Slow Voltage Variations

In radial MV networks such as the one considered here, the voltage variation in each line section can be approximated by:

$$\varepsilon_{m-n} = \frac{\Delta V}{V_n} \cdot 100\% \approx \frac{100\%}{V_n^2} (R_{m-n} P_{m-n} + X_{m-n} Q_{m-n}) \quad (44)$$

where:

- $V_n = 20 \text{ kV}$: nominal voltage
- R_{m-n}, X_{m-n} : resistance and reactance of section $m-n$
- P_{m-n}, Q_{m-n} : active and reactive power flowing through the section.

Positive power flow P, Q is considered from Node 3 toward the MV transformer bus, which is referred to as the reference node (Node 0), since its voltage is assumed to remain constant at 20 kV.

Line Data

For an ACSR-95 conductor:

$$R = 0.22 \text{ } \Omega/\text{km}, \quad X = 0.33 \text{ } \Omega/\text{km}. \quad (45)$$

Therefore, for line lengths $\ell_{01} = 3 \text{ km}$, $\ell_{12} = 8 \text{ km}$, and $\ell_{23} = 4 \text{ km}$, respectively:

- $R_{01} = 0.66 \text{ } \Omega$, $X_{01} = 0.99 \text{ } \Omega$,
- $R_{12} = 1.76 \text{ } \Omega$, $X_{12} = 2.64 \text{ } \Omega$,
- $R_{23} = 0.88 \text{ } \Omega$, $X_{23} = 1.32 \text{ } \Omega$

Selection of Worst-Case Scenarios

For the voltage-limit check, the two worst-case conditions for voltage rise and voltage drop are examined:

- **Min L – Max G**, i.e., minimum load and maximum generation, corresponding to the maximum voltage rise.
- **Max L – Min G**, i.e., maximum load and minimum generation, corresponding to the maximum voltage drop.

To assess slow voltage variation, the following two criteria are used:

$$0.95 V_n \leq \bar{V} \leq 1.05 V_n \implies |\bar{\epsilon}| = \frac{\epsilon_{\max} + \epsilon_{\min}}{2} \leq 5\% \quad (46)$$

$$|\Delta V| \leq 3\% V_n \implies |\delta\epsilon| = \frac{\epsilon_{\max} - \epsilon_{\min}}{2} \leq 3\% \quad (47)$$

Min L - Max G (Maximum Voltage Rise)

The following three tables present the calculations performed for this scenario. Specifically, Table 4 presents the load and PV generation at each node under **minimum load - maximum generation (Min L - Max G)** conditions. Table 5 presents the subsequent calculations for the power flow along each network path, together with the corresponding percentage voltage variation calculated using equation (44). Finally, Table 6 presents the cumulative voltage deviation at each node for this scenario.

Node	P_L (MW)	P_G (MW)	Q_L (MVar)	Q_G (MVar)
0	-5.000	+27.000	-2.130	0.00
1	-3.000	+5.400	-1.278	0.00
2	-1.000	+2.700	-0.426	0.00
3	0	+2.920	0.00	0.00

Table 4: Minimum load - Maximum generation power per node

Path	P_{m-n} (MW)	Q_{m-n} (MVar)	$R_{m-n} + jX_{m-n}$ (Ω)	ϵ_{m-n} (%)
$0 \rightarrow 1$	$P_{01} = P_1 + P_2 + P_3 = 7.020$	$Q_{01} = Q_1 + Q_2 + Q_3 = -1.704$	$0.66 + j0.99$	$\epsilon_{01} = 1.580$
$1 \rightarrow 2$	$P_{12} = P_2 + P_3 = 4.620$	$Q_{12} = Q_2 + Q_3 = -0.426$	$1.76 + j2.64$	$\epsilon_{12} = 1.752$
$2 \rightarrow 3$	$P_{23} = P_3 = 2.920$	$Q_{23} = Q_3 = 0$	$0.88 + j1.32$	$\epsilon_{23} = 0.642$

Table 5: Power flows, line impedances and voltage variation per path

Node	$\epsilon_{0 \rightarrow i}$ (%)
1	$\epsilon_{0 \rightarrow 1} = \epsilon_{01} = 1.580$
2	$\epsilon_{0 \rightarrow 2} = \epsilon_{01} + \epsilon_{12} = 3.332$
3	$\epsilon_{0 \rightarrow 3} = \epsilon_{01} + \epsilon_{12} + \epsilon_{23} = 3.974$

Table 6: Total voltage deviation up to node i

Max L - Min G (Maximum Voltage Drop)

Similarly, the following three tables present the calculations for this scenario. Specifically, Table 7 presents the load and PV generation at each node under **maximum load - minimum generation (Max L - Min G)** conditions. Table 8 presents the calculations for the power flow along each network path, together with the corresponding percentage voltage drop obtained from equation (44). Finally, Table 9 presents the cumulative voltage deviation at each node for this scenario.

Node	P_L (MW)	P_G (MW)	Q_L (MVar)	Q_G (MVar)
0	-20.000	0.000	-8.520	0.00
1	-7.000	0.000	-2.982	0.00
2	-3.000	0.000	-1.278	0.00
3	0	0.000	0.00	0.00

Table 7: Maximum load - Minimum generation power per node

Path	P_{m-n} (MW)	Q_{m-n} (MVar)	$R_{m-n} + jX_{m-n}$ (Ω)	ε_{m-n} (%)
$0 \rightarrow 1$	$P_{01} = P_1 + P_2 + P_3 = -10$	$Q_{01} = Q_1 + Q_2 + Q_3 = -4.260$	$0.66 + j0.99$	$\varepsilon_{01} = -2.704$
$1 \rightarrow 2$	$P_{12} = P_2 + P_3 = -3$	$Q_{12} = Q_2 + Q_3 = -1.278$	$1.76 + j2.64$	$\varepsilon_{12} = -2.163$
$2 \rightarrow 3$	$P_{23} = P_3 = 0$	$Q_{23} = Q_3 = 0$	$0.88 + j1.32$	$\varepsilon_{23} = 0.000$

Table 8: Power flows, line impedances and voltage variation per path

Node	$\varepsilon_{0 \rightarrow i}$ (%)
1	$\varepsilon_{0 \rightarrow 1} = \varepsilon_{01} = -2.704$
2	$\varepsilon_{0 \rightarrow 2} = \varepsilon_{01} + \varepsilon_{12} = -4.867$
3	$\varepsilon_{0 \rightarrow 3} = \varepsilon_{01} + \varepsilon_{12} + \varepsilon_{23} = -4.867$

Table 9: Total voltage deviation up to node i

Calculation of Mean Value and Voltage Variation

Using the maximum voltage rise and maximum voltage drop values obtained from Tables 6 and 9, together with criteria (46) and (47), we check whether the mean voltage and voltage variation are acceptable, as shown in Table 10:

Node	$\bar{\varepsilon}$ (%)	$\delta\varepsilon$ (%)
1	-0.562	2.142
2	-0.768	4.100
3	-0.447	4.421

Table 10: Mean voltage and voltage variation of each node

From Table 10, it can be observed that all mean voltage values are acceptable, whereas the voltage variation at Nodes 2 and 3 exceeds the specified limits. **Therefore, the new PV plant cannot be connected to Node 3 due to the slow voltage variation criterion.**

i) Construction of a Dedicated Line from the MV Substation Bus

In this case, a dedicated line is constructed and the PV plant is no longer connected to Node 3 of Figure 17; instead, it is connected directly to the MV bus of the substation through a new 15 km line. In addition, because of the dedicated line, the acceptable voltage variation range given by equation (47) changes from 5% to 8%. Node 3 will still be used to denote the node to which the considered 4 MW PV plant is connected.

For a complete evaluation of the connection of the PV plant through a dedicated line, all three required checks of the problem statement would need to be repeated. However, since the nominal power of the PV plant is significantly smaller than the nominal power of the HV/MV transformer, it is considered evident that the criteria related to both the transformer thermal loading limit and the short-circuit level at the 20 kV bus remain satisfied. Therefore, only the slow voltage variation criterion is checked below, using the same methodology as in the previous question (without the maximum and minimum load tables, since the corresponding load conditions remain unchanged in this case).

Min L - Max G (Maximum Voltage Rise)

Path	P_{m-n} (MW)	Q_{m-n} (MVar)	$R_{m-n} + jX_{m-n}$ (Ω)	ε_{m-n} (%)
$0 \rightarrow 1$	$P_{01} = P_1 + P_2 = 4.100$	$Q_{01} = Q_1 + Q_2 = -1.704$	$0.66 + j0.99$	$\varepsilon_{01} = 0.255$
$1 \rightarrow 2$	$P_{12} = P_2 = 1.700$	$Q_{12} = Q_2 = -0.426$	$1.76 + j2.64$	$\varepsilon_{12} = 0.476$
$0 \rightarrow 3$	$P_{03} = P_3 = 2.92$	$Q_{03} = Q_3 = 0$	$3.30 + j4.95$	$\varepsilon_{03} = 2.409$

Table 11: Power flows, line impedances and voltage variation per path

Node	$\varepsilon_{0 \rightarrow i}$ (%)
1	$\varepsilon_{0 \rightarrow 1} = \varepsilon_{01} = 0.255$
2	$\varepsilon_{0 \rightarrow 2} = \varepsilon_{01} + \varepsilon_{12} = 0.722$
3	$\varepsilon_{0 \rightarrow 3} = 2.409$

Table 12: Total voltage deviation up to node i

Max L - Min G (Maximum Voltage Drop)

Path	P_{m-n} (MW)	Q_{m-n} (MVar)	$R_{m-n} + jX_{m-n}$ (Ω)	ε_{m-n} (%)
$0 \rightarrow 1$	$P_{01} = P_1 + P_2 = -10$	$Q_{01} = Q_1 + Q_2 = -4.260$	$0.66 + j0.99$	$\varepsilon_{01} = -2.704$
$1 \rightarrow 2$	$P_{12} = P_2 = -3$	$Q_{12} = Q_2 = -1.278$	$1.76 + j2.64$	$\varepsilon_{12} = -2.193$
$0 \rightarrow 3$	$P_{03} = P_3 = 0$	$Q_{03} = Q_3 = 0$	$3.30 + j4.95$	$\varepsilon_{03} = 0.000$

Table 13: Power flows, line impedances and voltage variation per path

Node	$\varepsilon_{0 \rightarrow i}$ (%)
1	$\varepsilon_{0 \rightarrow 1} = \varepsilon_{01} = -2.704$
2	$\varepsilon_{0 \rightarrow 2} = \varepsilon_{01} + \varepsilon_{12} = -4.867$
3	$\varepsilon_{0 \rightarrow 3} = 0$

Table 14: Total voltage deviation up to node i

Calculation of Mean Value and Voltage Variation

As in the previous question, using the maximum voltage rise and maximum voltage drop values obtained from Tables 12 and 14, together with criteria (46) and (47), we check whether the mean voltage and voltage variation are acceptable (for Node 3, the allowable voltage variation is 8%), as shown in Table 15:

Node	$\bar{\epsilon}(\%)$	$\delta\epsilon(\%)$
1	-1.225	1.480
2	-2.073	2.795
3	1.205	1.205

Table 15: Mean voltage and voltage variation of each node

From Table 15, it can be observed that all mean-voltage values and voltage-variation values are acceptable. **Therefore, the new PV plant can be connected to Node 3 through a dedicated line.**

ii) Co-Location of a Battery Energy Storage System

In this case, a battery energy storage system is added at Node 3 in order to reduce the voltage fluctuation observed at that node. Specifically, the battery will absorb its maximum power during maximum generation and inject its maximum power during maximum load, thereby compensating for the voltage rise and voltage drop at the node, respectively.

The slow voltage variation analysis presented in the previous questions is repeated below. To determine the minimum battery power rating, we denote it by $\mathbf{x}$.

Min L - Max G (Maximum Voltage Rise)

Node	P_L (MW)	P_G (MW)	Q_L (MVar)	Q_G (MVar)
0	-5.000	+27.000	-2.130	0.00
1	-3.000	+5.400	-1.278	0.00
2	-1.000	+2.700	-0.426	0.00
3	-x	+2.920	0.00	0.00

Table 16: Minimum load - Maximum generation power per node

Path	P_{m-n} (MW)	Q_{m-n} (MVar)	$R_{m-n} + jX_{m-n}$ (Ω)	ϵ_{m-n} (%)
0 $\rightarrow$ 1	$P_{01} = P_1 + P_2 + P_3 = 7.020 - \mathbf{x}$	$Q_{01} = Q_1 + Q_2 + Q_3 = -3.840$	$0.66 + j0.99$	$\epsilon_{01} = 0.208 - 0.165\mathbf{x}$
1 $\rightarrow$ 2	$P_{12} = P_2 + P_3 = 4.620 - \mathbf{x}$	$Q_{12} = Q_2 + Q_3 = -0.426$	$1.76 + j2.64$	$\epsilon_{12} = 1.752 - 0.44\mathbf{x}$
2 $\rightarrow$ 3	$P_{23} = P_3 = 2.920 - \mathbf{x}$	$Q_{23} = Q_3 = 0$	$0.88 + j1.32$	$\epsilon_{23} = 0.642 - 0.22\mathbf{x}$

Table 17: Power flows, line impedances and voltage variation per path

Node	$\epsilon_{0 \rightarrow i}$ (%)
1	$\epsilon_{0 \rightarrow 1} = \epsilon_{01} = 0.208 - 0.165\mathbf{x}$
2	$\epsilon_{0 \rightarrow 2} = \epsilon_{01} + \epsilon_{12} = 1.960 - 0.605\mathbf{x}$
3	$\epsilon_{0 \rightarrow 3} = \epsilon_{01} + \epsilon_{12} + \epsilon_{23} = 2.602 - 0.825\mathbf{x}$

Table 18: Total voltage deviation up to node i

Max L - Min G (Maximum Voltage Drop)

Node	P_L (MW)	P_G (MW)	Q_L (MVar)	Q_G (MVar)
0	-20.000	0.000	-8.520	0.00
1	-7.000	0.000	-2.982	0.00
2	-3.000	0.000	-1.278	0.00
3	0	$\mathbf{x}$	0.00	0.00

Table 19: Maximum load - Minimum generation power per node

Path	P_{m-n} (MW)	Q_{m-n} (MVar)	$R_{m-n} + jX_{m-n}$ (Ω)	ε_{m-n} (%)
0 $\rightarrow$ 1	$P_{01} = P_1 + P_2 + P_3 = -10 + \mathbf{x}$	$Q_{01} = Q_1 + Q_2 + Q_3 = -4.260$	$0.66 + j0.99$	$\varepsilon_{01} = -2.704 + 0.165\mathbf{x}$
1 $\rightarrow$ 2	$P_{12} = P_2 + P_3 = -3 + \mathbf{x}$	$Q_{12} = Q_2 + Q_3 = -1.278$	$1.76 + j2.64$	$\varepsilon_{12} = -2.163 + 0.44\mathbf{x}$
2 $\rightarrow$ 3	$P_{23} = P_3 = \mathbf{x}$	$Q_{23} = Q_3 = 0$	$0.88 + j1.32$	$\varepsilon_{23} = 0.22\mathbf{x}$

Table 20: Power flows, line impedances and voltage variation per path

Node	$\varepsilon_{0 \rightarrow i}$ (%)
1	$\varepsilon_{0 \rightarrow 1} = \varepsilon_{01} = -2.704 + 0.165\mathbf{x}$
2	$\varepsilon_{0 \rightarrow 2} = \varepsilon_{01} + \varepsilon_{12} = -4.867 + 0.605\mathbf{x}$
3	$\varepsilon_{0 \rightarrow 3} = \varepsilon_{01} + \varepsilon_{12} + \varepsilon_{23} = -4.867 + 0.825\mathbf{x}$

Table 21: Total voltage deviation up to node i

Calculation of Minimum Battery Power

Using the maximum voltage rise and maximum voltage drop values obtained from Tables 18 and 21, the mean voltage and voltage variation are shown in Table 22.

Node	$\bar{\varepsilon}$ (%)	$\delta\varepsilon$ (%)
1	-1.248	$1.456 - 0.165\mathbf{x}$
2	-1.144	$3.667 - 0.605\mathbf{x}$
3	-1.133	$3.998 - 0.825\mathbf{x}$

Table 22: Mean voltage and voltage variation of each node

Applying criteria (46) and (47), we obtain the following two equations for Nodes 2 and 3, respectively:

$$3.667 - 0.605\mathbf{x} \leq 3 \implies \mathbf{x} \leq 1.102 \text{ MW} \quad (48)$$

$$3.998 - 0.825\mathbf{x} \leq 3 \implies \mathbf{x} \leq 1.210 \text{ MW} \quad (49)$$

Therefore, the minimum battery energy storage system power rating that makes the connection of the PV plant at Node 3 acceptable is 1.102 MW.

At this point, it is worth noting that the required battery power is very small compared with the short-circuit level limit of the bus and the transformer thermal limit, so re-evaluating these criteria is unnecessary.

PROBLEM 3: ENERGY YIELD ESTIMATION & ECONOMIC EVALUATION

1. Daily Market Price and Residual Load Curves

To evaluate the operation of the electricity market, the daily curves of the average day-ahead market clearing price and the average residual load of the system were calculated.

The residual load is defined as:

$$P_{\text{residual}} = P_L - P_{\text{RES}} \quad (50)$$

For the calculation of the daily curves, the average value of the above quantities was calculated for each hour over the entire year, and the results are presented in Figure 18 using different vertical-axis scales.

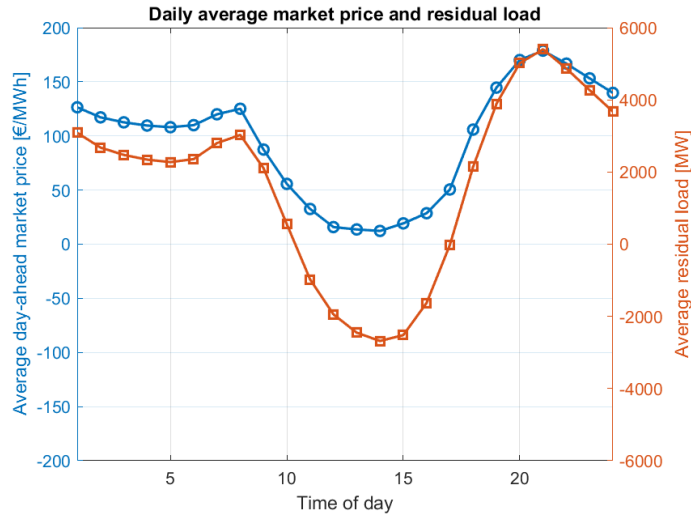


Figure 18: Daily curve of average market price and average residual load

It can be observed that the average day-ahead market clearing price is directly correlated with the variation of the residual load. This trend is expected, since increased RES penetration, which provides lower-cost energy, leads to a reduction in residual load and consequently to a decrease in the system marginal price. It is also noteworthy that during midday hours, when RES generation peaks, the residual load becomes negative, indicating power exports to neighboring systems.

2. Annual Energy Production, Capacity Factor, and PV Revenues

To estimate the expected annual energy production, capacity factor, and revenues of the PV plant from the day-ahead market, the following procedure is applied.

To calculate the injected power of the PV plant, the main operating constraints of the electrical system are taken into account. Specifically, a minimum thermal generation of $P_{\text{th,min}} = 500$ MW is imposed

for reserve requirements, while the maximum possible power export to neighboring systems depends on the time of day and is given by:

$$P_{\text{exp,max}} = \begin{cases} 2500 \text{ MW,} & \text{for all other hours} \\ 1750 \text{ MW,} & \text{for the hours 08:00–11:00 and 14:00–17:00} \\ 1000 \text{ MW,} & \text{for the hours 11:00–14:00} \end{cases} \quad (51)$$

During hours in which a power surplus occurs that cannot be absorbed by the system, i.e., when:

$$P_{\text{RES,tot}} + P_{\text{th,min}} > P_{\text{Load}} + P_{\text{exp,max}} \quad (52)$$

the required rejected power is calculated as:

$$P_{\text{rejected}} = P_{\text{RES,tot}} + P_{\text{th,min}} - P_{\text{Load}} - P_{\text{exp,max}}, \quad P_{\text{rejected}} \geq 0 \quad (53)$$

A proportional curtailment of RES generation is assumed, so the rejected power corresponding to the PV plant is:

$$P_{\text{PV,rejected}} = P_{\text{rejected}} \cdot \frac{P_{\text{PV}}}{P_{\text{RES,tot}}} \quad (54)$$

The final injected power of the PV plant into the grid, before applying the connection limit, is:

$$P_{\text{PV,left}} = P_{\text{PV}} - P_{\text{PV,rejected}} \quad (55)$$

According to the problem statement, a maximum connection-power limit equal to 72% of the nominal power of the PV plant is applied ($P_{\text{nom}} = 30 \text{ MW}$), so the final injected power is:

$$P_{\text{PV,inj}} = \min(P_{\text{PV,left}}, 0.72 \cdot P_{\text{nom}}) = \min(P_{\text{PV,left}}, 21.6) \quad (56)$$

The expected annual energy production of the PV plant is:

$$E_{\text{PV}} = \sum_{h=1}^{8760} P_{\text{PV,inj}} \quad (57)$$

The capacity factor of the PV plant is:

$$CF = \frac{E_{\text{PV}}}{P_{\text{nom}} \cdot 8760} \cdot 100\% \quad (58)$$

The annual revenues of the PV plant from participation in the day-ahead market are calculated by:

$$R = \sum_{h=1}^{8760} P_{\text{PV,inj}} \cdot \text{DAM} \quad (59)$$

The average selling price of the generated energy is given by:

$$\bar{\lambda} = \frac{R}{E_{\text{PV}}} \quad (60)$$

Finally, the percentage of PV energy rejected due to the inability of the system to absorb it is:

$$\% \text{Rejected} = \frac{\sum_{h=1}^{8760} P_{\text{PV,rejected}}}{\sum_{h=1}^{8760} P_{\text{PV}}} \cdot 100\% \quad (61)$$

The results are summarized in Table 23, which presents the annual energy production, capacity factor, PV revenues from the day-ahead market, and the percentage of rejected energy due to system constraints.

E_{PV} [MWh]	42,644.69
CF [%]	16.23
R [€]	1,205,026.44
λ [€/MWh]	28.26
Rejected [%]	15.59

Table 23: Summary of results regarding the PV plant

3. Effect of a 15 MWh Battery Energy Storage System on the PV Plant

In this question, the annual injected energy, capacity factor, and revenues of the PV plant are recalculated for the case of co-locating a battery energy storage system with maximum power $P_{bat,max} = 15$ MW and energy capacity $E_{bat,max} = 15$ MWh. The battery is charged exclusively from PV generation; therefore, battery operation is applied to the available PV power before system constraints are imposed.

The battery operation is defined as follows:

- During the period 13:00–14:00, the battery is charged by absorbing power up to 15 MW, exclusively from the PV plant.
- During the period 20:00–21:00, the battery is discharged and delivers to the grid the energy stored during the same day.
- The stored energy cannot exceed the capacity of 15 MWh.
- The round-trip charging–discharging efficiency is taken as $\eta_{rt} = 0.85$, and energy transfer from/to another day is not allowed.

The charging energy during the interval 13:00–14:00 is limited by the battery power rating, battery capacity, and available PV power:

$$E_{bat,ch} = \min(P_{bat,max}, E_{bat,max}, P_{PV}^{13:00-14:00}) \quad (62)$$

The discharged energy during 20:00–21:00 is:

$$E_{bat,dis} = \eta E_{bat,ch}, \quad \eta_{rt} = 0.85 \quad (63)$$

The net PV power available for injection is:

$$P_{PV,net} = P_{PV} - P_{bat,ch}^{13:00-14:00} + P_{bat,dis}^{20:00-21:00} \quad (64)$$

Then, $P_{PV,net}$ is subjected to the same system operating constraints as in Question 2. The same procedure is therefore followed, using equations (52) through (61), resulting in the values shown in Table 24.

E_{PV} [MWh]	43,067.43
CF [%]	16.39
R [€]	1,884,725.97
λ [€/MWh]	43.76
Rejected [%]	13.47

Table 24: Summary of results regarding the PV plant using batteries

Comparing the results of Question 2 with the case including battery storage, an overall improvement in all key performance indicators of the PV plant can be observed. Annual injected energy, capacity factor, and revenues all increase, indicating that the PV plant is overall more efficient. Particular attention should be paid to the average selling price of the generated energy, which is significantly higher than in the case without battery storage. This increase is expected because the stored energy is injected into the grid during the evening peak-demand hours, when market prices are higher.

4. Calculation of the Minimum Bid Price Using LCOE

In this question, the minimum bid price (€/MWh) that could be submitted by the producer in a competitive auction is calculated. This price corresponds to the Levelized Cost of Energy (*LCOE*) of the investment and is obtained from the ratio of the discounted total cost to the discounted energy produced during the lifetime of the project. The cost data for the considered project are presented in Table 25.

PV plant investment cost	500 €/kW
Battery system investment cost	120 €/kWh
Evaluation period	20 years
Annual operation and maintenance cost	2.5% of total investment cost
RES levy (Art. 87, Law 4964/2022)	2 €/MWh
Discount rate	8%
Debt financing	65% of investment cost
Loan interest rate	5%
Loan repayment period	15 years
Depreciation of fixed assets	20 years – straight-line
Corporate tax rate	22%
Annual inflation rate (applied to operating costs)	2%

Table 25: Cost data for the PV plant and the battery energy storage system

For the LCOE calculation, the annual energy production of the PV plant is assumed to remain constant throughout the evaluation period, since PV performance degradation over time is neglected. The general LCOE expression is:

$$LCOE = \frac{K_{0,tot} + \sum_{t=1}^N \frac{-A_t \cdot \Phi\Sigma + \Lambda\Delta_t \cdot (1 - \Phi\Sigma)}{(1 + k_{tot})^t} - \frac{Y_{A_N}}{(1 + k_{tot})^N}}{(1 - \Phi\Sigma) \cdot \sum_{t=1}^N \frac{E_{gen,t}}{(1 + k_{tot})^t}} \quad (65)$$

where:

- $E_{gen} = 42,644.69$ MWh/year: Annual energy generation, as calculated in Question 2
- $N = 20$ years: Investment lifetime in years
- $\Phi\Sigma = 22\%$: Corporate tax rate used for income tax calculations
- $Y_{A_N} = 0$: Residual value of the investment at the end of the evaluation period

- $K_{0,\text{tot}} = 15 \text{ M€}$: Total initial investment cost:

$$K_{0,\text{tot}} = 500 \text{ €/kW} \cdot 30,000 \text{ kW} = 15,000,000 \text{ €} = 15 \text{ M€} \quad (66)$$

- $A_t = 0.75 \text{ M€/year}$: Annual depreciation:

$$A_t = \frac{K_{0,\text{tot}}}{N} = \frac{15 \text{ M€}}{20} = 0.75 \text{ M€/year} \quad (67)$$

- $\Lambda\Delta_t$: Annual operating expenses, consisting of operation and maintenance costs and the RES levy. The operation and maintenance cost is calculated as:

$$0.025 \cdot K_{0,\text{tot}} \cdot (1 + 0.02)^t = 0.025 \cdot 15,000,000 \cdot (1 + 0.02)^t = 375,000 \cdot (1 + 0.02)^t \text{ €} \quad (68)$$

The RES levy is:

$$2 \cdot E_{\text{gen}} = 2 \cdot 42,644.69 = 85,289.38 \text{ €/year} \quad (69)$$

Therefore, the total annual operating expenses are:

$$\Lambda\Delta_t = 375,000 \cdot (1 + 0.02)^t + 85,289.38 \text{ €} \quad (70)$$

- k_{tot} : The weighted average cost of debt and equity, calculated as:

$$k_{\text{tot}} = P_d k_d(1 - \Phi\Sigma) + P_{eq} k_{eq} = 0.65 \cdot 0.05 \cdot (1 - 0.22) + 0.35 \cdot 0.08 = 5.335\% \quad (71)$$

where:

- $P_d = 65\%$: Debt share in the total investment
- $k_d = 5\%$: Borrowing interest rate
- $P_{eq} = 35\%$: Equity share in the total investment
- $k_{eq} = 8\%$: Cost of equity.

For the case without a battery system, the initial investment cost consists exclusively of the PV plant installation, while annual depreciation is considered linear. Operating expenses include operation and maintenance costs, increased annually due to inflation, as well as the RES levy, which is proportional to the annual energy produced.

Using the above calculations and equation (65), we obtain:

$$LCOE = 45.79 \text{ €/MWh} \quad (72)$$

This value corresponds to the minimum bid price that could be submitted in a competitive auction for the investment to be marginally viable, i.e., to yield neither profit nor loss.

In the second case, where the PV plant is co-located with a battery system, the initial investment cost increases due to the additional capital cost of the storage system and becomes:

$$K_{0,\text{tot}} = 500 \text{ €/kW} \cdot 30,000 \text{ kW} + 120 \text{ €/kW} \cdot 15,000 \text{ kW} = 16.8 \text{ M€} \quad (73)$$

Along with the total installation cost, the operating expenses and projected depreciation also change. At the same time, annual energy production changes, and the energy is sold at a different time of day, thereby changing the annual revenue level while taking into account the round-trip charging–discharging efficiency.

Following the same methodology, the LCOE for the case with the battery system is:

$$LCOE_{\text{batt}} = 49.46 \text{ €/MWh} \quad (74)$$

It can be observed that although a significant increase in the PV plant revenues was calculated in the previous question, the increase in total investment cost leads to a higher levelized cost of energy. Therefore, in this case, battery co-location does not reduce the minimum required bid price in a competitive auction, making the option without battery installation economically preferable.

5. Economic Evaluation of the Investment with Government Support

For the economic evaluation of an investment, several financial indicators can be used, with the most common being Net Present Value (NPV), Internal Rate of Return (IRR), and Discounted Payback Period (DPP). These indicators are used to assess the viability and profitability of an investment.

In this question, it is assumed that the plant receives government support equal to 95 €/MWh during the hours in which the day-ahead market clearing price is positive, while no support is provided during zero-price hours. Therefore, the effective market clearing price is:

$$DAM_h^{\text{new}} = \begin{cases} 95, & \text{when } DAM_h > 0 \\ 0, & \text{when } DAM_h \leq 0 \end{cases} \quad (75)$$

The annual revenues of the plant from electricity sales are calculated as:

$$E_t = \sum_{h=1}^{8760} P_{\text{PV},\text{inj},h} \cdot DAM_h^{\text{new}} \quad (76)$$

The annual debt service is calculated from:

$$\Delta\Delta_t = \left(k_d + \frac{k_d}{(1 + k_d)^{N_d} - 1} \right) \cdot K_d \quad (77)$$

where:

- $k_d = 5\%$: Borrowing interest rate
- $N_d = 15 \text{ years}$: Loan repayment period in years
- $K_d = 0.65 \cdot K_{0,\text{tot}}$: Debt capital

Assuming constant principal repayment, the principal amount repaid in year t is calculated as:

$$X_t = \frac{k_d}{(1 + k_d)^{N_d} - 1} K_d (1 + k_d)^{t-1} \quad (78)$$

Annual interest can then be obtained from:

$$T_t = \Delta\Delta_t - X_t \quad (79)$$

Corporate income tax is calculated from taxable income as:

$$\Phi_t = \max(0, E_t - \Lambda\Delta_t - A_t - T_t) \cdot \Phi\Sigma \quad (80)$$

where:

- $\Phi\Sigma = 22\%$: Corporate tax rate
- $A_t = 0.75 \text{ M€}/\text{year}$ or $0.84 \text{ M€}/\text{year}$: Annual depreciation of the investment (without and with the battery system)
- $\Lambda\Delta_t$: Annual operating expenses of the plant

The annual Net Cash Flow (NCF), using the above quantities and evaluated from the equity perspective, is defined as:

$$NCF_t = E_t - \Lambda\Delta_t - \Phi_t - \Delta\Delta_t = (E_t - \Lambda\Delta_t - A_t - T_t) \cdot (1 - \Phi\Sigma) + A_t - X_t \quad (81)$$

The Net Present Value (NPV) of the investment is defined as the value of the investment at the beginning of commercial operation and is given by:

$$NPV = -K_{0,eq} + \sum_{t=1}^N \frac{NCF_t}{(1 + k_{eq})^t} + \frac{Y_{A_N}}{(1 + k_{eq})^N} \quad (82)$$

where:

- $K_{0,eq} = 0.35 \cdot K_{0,tot}$: Equity investment
- $k_{eq} = 8\%$: Discount rate, representing the required return on equity
- $N = 20$ years: Investment lifetime in years
- $Y_{A_N} = 0$: Residual value of the investment in year N

The Internal Rate of Return (IRR) is defined as the discount rate that makes the Net Present Value (NPV) equal to zero, obtained from:

$$-K_{0,eq} + \sum_{t=1}^N \frac{NCF_t}{(1 + IRR)^t} = 0 \quad (83)$$

The Discounted Payback Period (DPP) is defined as the number of years (x) required for the cumulative discounted Net Cash Flow (NCF) to recover the initial investment cost:

$$-K_{0,eq} + \sum_{t=1}^x \frac{NCF_t}{(1 + k_{eq})^t} \geq 0 \quad x \in \mathbb{Z} \quad (84)$$

The final results for Net Present Value (NPV), Internal Rate of Return (IRR), and Discounted Payback Period (DPP) are presented in Table 26, both with and without battery co-location.

Financial Indicators	PV without batteries	PV with batteries
NPV [€]	532,523.71	1,110,302.60
IRR [%]	9.15	10.12
DPP [years]	19	17

Table 26: Summary of financial evaluation results

The results of the economic evaluation show that the integration of a battery energy storage system improves all major financial indicators. This improvement is attributed to the ability to shift energy production to periods with higher market prices, which increases the annual revenues of the plant and makes the investment more attractive, even though the LCOE analysis showed that co-location leads to a higher energy production cost per unit of energy. This difference is mainly due to the difference in energy selling prices under the government support scheme.

PROBLEM 4: SIMULTANEOUS COMPENSATION (NET BILLING)

1. Monthly and Annual Electricity Cost

A household consumer is considered with monthly electricity consumption E_m (kWh), according to Table 27, and contracted apparent power $S_{nom} = 8$ kVA. The calculations are based on the tariff shown in Figure 19, without considering other charges (fixed charges, VAT, municipal fees, etc.).

Month	Consumption (kWh)	PV utilization factor (%)	Injected energy value (€/kWh)
1	530	12	0.09
2	440	13	0.07
3	390	17	0.04
4	310	20	0.03
5	330	21	0.02
6	520	23	0.05
7	670	23	0.06
8	570	22	0.05
9	410	20	0.06
10	370	16	0.06
11	440	12	0.04
12	520	11	0.08

Table 27: Monthly electricity consumption, PV utilization factor and injected energy value

Προμήθεια (€/kWh)	Ρυθμιζόμενες Χρεώσεις					
	Σύστημα Μεταφοράς (ΧΧΣ)		Δίκτυο Διανομής (ΧΧΔ)		Λοιπές Χρεώσεις	ΕΤΜΕΑΡ
	Υπόλοιπο (€/kVA·ΣΙ·ημέρες/365)	Ενέργεια (€/kWh)	Υπόλοιπο (€/kVA·ΣΙ·ημέρες/365)	Ενέργεια (€/kWh)	(€/kWh)	ΥΚΩ
0,14	-	0,00999	6,210	0,00339	0,00007	0,017

Φόροι	
ΕΦΚ	Ειδικό Τέλος 5%
(€/kWh)	(Χρέωση προμήθειας + Ρυθμιζόμενες Χρεώσεις - ΕΤΜΕΑΡ + ΕΦΚ)* 5%
0,0022	

Figure 19: Household consumer invoice

For the calculations, each month is assumed to have 30 days and the household power factor is assumed equal to 1.

The Distribution Network Capacity Charge (XXD) for month m , with nominal power $P_{\text{nom}} = 8 \text{ kW}$, is calculated as:

$$C_{\text{dist,P},m} = 6.210 \cdot P_{\text{nom}} \cdot \frac{30}{365} = 6.210 \cdot 8 \cdot \frac{30}{365} = 4.083 \text{ €} \quad (85)$$

The special 5 levy for month m , where E_m is the monthly electricity consumption, is:

$$\begin{aligned} C_{\text{spec},m} &= 0.005 \cdot (E_m \cdot (0.14 + 0.00999 + 0.00339 + 0.00007 + 0.017 + 0.0069) + C_{\text{dist,P},m}) \\ &= 0.005 \cdot (0.17735 \cdot E_m + 4.083) \\ &= 0.00088675 \cdot E_m + 0.020415 \end{aligned} \quad (86)$$

Therefore, the total monthly cost C_m is:

$$\begin{aligned} C_m &= E_m \cdot 0.17735 + C_{\text{dist,P},m} + C_{\text{spec},m} \\ &= 0.17735 \cdot E_m + 4.083 + (0.00088675 \cdot E_m + 0.020415) \\ &= 0.17823675 \cdot E_m + 4.103415 \end{aligned} \quad (87)$$

The electricity cost for each month before PV installation is presented in Table 28:

Month	Consumption (kWh)	Cost (€)
1	530	99.65
2	440	83.43
3	390	74.41
4	310	59.99
5	330	63.60
6	520	97.85
7	670	124.89
8	570	106.86
9	410	78.02
10	370	70.81
11	440	83.43
12	520	97.85
Total	5500	1040.77

Table 28: Monthly electricity cost before PV installation.

The annual electricity cost before PV installation is:

$$C_{\text{year}} = \sum_{m=1}^{12} C_m = 1040.77 \text{ €} \quad (88)$$

The average monthly electricity bill before PV installation is:

$$\bar{C}_m = \frac{C_{\text{year}}}{12} = 86.725 \text{ €} \quad (89)$$

2. Monthly and Annual Electricity Cost with PV

After installing a PV system with power $P_{PV} = 5 \text{ kW}$, the monthly PV production is given by the following equation, where UV_m is the utilization factor:

$$E_{PV,m} = P_{PV} (24 \cdot 30) UV_m = 5 \cdot 24 \cdot 30 \cdot UV_m = 3600 \cdot UV_m \quad (90)$$

The production-consumption coincidence factor is equal to 25%, therefore the energy directly consumed by the household is:

$$E_{\text{self},m} = \min(0.25 \cdot E_{PV,m}, E_{\text{load}}) \quad (91)$$

The energy injected into the grid is:

$$E_{\text{inj},m} = E_{\text{self},m} - E_{\text{load}} \quad (92)$$

Therefore, the energy drawn from the grid is:

$$E_{\text{grid},m} = E_{\text{load},m} - E_{\text{self},m} \quad (93)$$

The charges (supply, regulated charges, and taxes) are calculated as in Question 1, replacing the consumption E_m with the grid-imported energy $E_{\text{grid},m}$.

The value of the injected energy is calculated using the IEV_m value given in Table 27:

$$R_m = IEV_m \cdot E_{\text{inj},m} \quad (94)$$

Therefore, the consumer's net monthly amount is:

$$C_m^{\text{net}} = C_m^{\text{gross}} - R_m \quad (95)$$

The monthly electricity cost after PV installation is presented in Table 29:

Month	E_{grid} (kWh)	E_{inj} (kWh)	E_{self} (kWh)	Credit (€)	Cost (€)
1	422.0	324.0	108.0	29.16	51.02
2	323.0	351.0	117.0	24.57	37.76
3	237.0	459.0	153.0	18.36	28.47
4	130.0	540.0	180.0	16.20	11.34
5	141.0	567.0	189.0	11.34	18.18
6	313.0	621.0	207.0	31.05	29.48
7	463.0	621.0	207.0	37.26	50.31
8	372.0	594.0	198.0	29.70	41.47
9	230.0	540.0	180.0	32.40	13.17
10	226.0	432.0	144.0	25.92	18.93
11	332.0	324.0	108.0	12.96	51.00
12	421.0	297.0	99.0	23.76	56.24
Total	3610.0	5670.0	1890.0	292.68	407.37

Table 29: Monthly billing after PV installation

The annual electricity cost after PV installation is:

$$C_{\text{year}}^{\text{net}} = \sum_{m=1}^{12} C_m^{\text{net}} = 407.37 \text{ €} \quad (96)$$

The average monthly electricity bill after PV installation is:

$$\bar{C}_m = \frac{C_{\text{year}}}{12} = 33.95 \text{ €} \quad (97)$$

Therefore, the annual consumer savings are:

$$\Delta C = 1040.77 - 407.37 = 633.40 \text{ €} \quad (98)$$

3. Payback Period and Net Present Value of the PV Investment

The cost parameters of the residential PV installation are presented in Table 30.

PV system cost (€/kWp)	1000
Grid connection cost (€)	600
Discount rate (%)	4
Investment lifetime (years)	25

Table 30: Cost parameters for the PV system installation.

The annual cash flow of the investment is taken to be the annual consumer savings due to the PV system installation, as calculated in Question 2, and is denoted by KTP_t .

The payback period of the investment is determined from the condition that the discounted value of the future cash flows equals the initial investment cost:

$$-K_{0,eq} + \sum_{t=1}^x \frac{KTP_t}{(1 + k_{eq})^t} \geq 0 \quad x \in \mathbb{Z} \quad (99)$$

where:

- $K_{0,eq} = 5 \cdot 1000 + 600 = 5600$ €: Equivalent initial investment cost,
- $KTP_t = 633.40$ €: Annual net cash flow from savings,
- $k_{eq} = 4\%$: Discount rate,
- x : Investment payback period in years.

The Net Present Value (NPV) of the investment over a lifetime of $N = 25$ years is calculated from:

$$NPV = -K_{0,eq} + \sum_{t=1}^N \frac{KTP_t}{(1 + k_{eq})^t} \quad (100)$$

Finally, it is calculated that the payback period of the PV installation investment for the residential consumer is **12 years**, while the Net Present Value (NPV) of the savings is 4295.10 €.

4. Battery Installation and Increase in Self-Consumption

The installation of a battery energy storage system in addition to the PV system is considered, increasing the coincidence of PV generation with household consumption from 25% to 60%.

The same procedure as in Question 2 is repeated, and the results are presented in Table 31.

Month	E_{grid} (kWh)	E_{inj} (kWh)	E_{self} (kWh)	Credit (€)	Cost (€)
1	270.80	172.80	259.20	15.55	37.37
2	159.20	187.20	280.80	13.10	19.70
3	22.80	244.80	367.20	9.79	-1.58
4	0.00	410.00	310.00	12.30	-8.20
5	0.00	426.00	330.00	8.52	-4.42
6	23.20	331.20	496.80	16.56	-8.27
7	173.20	331.20	496.80	19.87	15.46
8	94.80	316.80	475.20	15.84	5.35
9	0.00	310.00	410.00	18.60	-14.50
10	24.40	230.40	345.60	13.82	-5.32
11	180.80	172.80	259.20	6.91	29.79
12	282.40	158.40	237.60	12.67	42.34
Total	1231.60	3291.60	4268.40	163.53	107.72

Table 31: Monthly billing after PV installation with battery system.

The annual electricity cost with the PV and battery installation is:

$$C_{year}^{net} = \sum_{m=1}^{12} C_m^{net} = 107.72 \text{ €} \quad (101)$$

The average monthly electricity bill with the PV and battery installation is:

$$\bar{C}_m = \frac{C_{\text{year}}}{12} = 8.98 \text{ €} \quad (102)$$

Therefore, the annual consumer savings relative to the PV-only case are:

$$\Delta C = 407.37 - 107.72 = 299.65 \text{ €} \quad (103)$$

5. Maximum Acceptable Battery Installation Cost

The investment decision criterion for the battery installation is the Net Present Value (NPV) of the additional savings relative to the PV-only case.

The maximum acceptable battery installation cost is obtained from the zero-NPV condition, using as annual cash flow the additional savings due to battery installation, equal to $KTP_t = \Delta C = 299.64 \text{ €/year}$:

$$\text{NPV} = 0 \implies K_{0,\text{eq},\text{lim}} = \sum_{t=1}^N \frac{KTP_t}{(1 + k_{\text{eq}})^t} \quad (104)$$

Therefore, based on the above relation, the maximum battery installation cost that can be borne by the consumer is 4681.01 €.